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1 Question 1

The window parameter w determines the scope of the "context" that the algorithm considers when learning the representation of a specific node. Conceptually, it controls the trade-off between capturing **local** versus **global** structural information.

When w is chosen to be very small, the algorithm restricts its focus to immediate neighbors. Consequently, the resulting embeddings strongly emphasize homophily (microscopic structure), meaning that nodes are mapped close together in the embedding space primarily because they are directly connected or share immediate neighbors.

Conversely, choosing a very large w allows the algorithm to "look" much further along the random walk, capturing higher-order proximity and broader graph topology. This shifts the emphasis toward structural equivalence (macroscopic structure); nodes may receive similar embeddings not because they are close to each other, but because they play similar roles within the graph (hubs or bridges) or belong to the same large community. However, if w is too large, the embeddings may become overly smoothed, blurring the distinct local features that differentiate specific nodes.

2 Question 2

Comparing the two embedding matrices $\mathbf{X}_1$ and $\mathbf{X}_2$, we can observe a specific linear relationship between them.
By inspecting the columns of the two matrices, we can see that:

- The first column of $\mathbf{X}_2$ is identical to the first column of $\mathbf{X}_1$.
- The second column of $\mathbf{X}_2$ is exactly the negation of the second column of $\mathbf{X}_1$ (the signs are flipped).

Mathematically, $\mathbf{X}_2$ can be obtained from $\mathbf{X}_1$ by right-multiplying with a reflection matrix $\mathbf{Q}$:

$$\mathbf{X}_2 = \mathbf{X}_1\mathbf{Q} \quad \text{where} \quad \mathbf{Q} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Geometrically, the set of points represented by $\mathbf{X}_2$ is a **reflection** of the points in $\mathbf{X}_1$ across the first dimension (the x-axis). Since $\mathbf{Q}$ is an orthogonal matrix (specifically, $\mathbf{Q}^T\mathbf{Q} = \mathbf{I}$), this transformation is isometric, meaning it preserves the distances and angles between all pairs of nodes.

In the context of DeepWalk, the absolute coordinates of the embeddings are not unique. These algorithms aim to preserve relative similarities, typically measured by the dot product $\langle \mathbf{u}, \mathbf{v} \rangle$.

Since orthogonal transformations preserve the dot product:

$$\langle \mathbf{uQ}, \mathbf{vQ} \rangle = (\mathbf{uQ})(\mathbf{vQ})^T = \mathbf{uQQ}^T\mathbf{v}^T = \mathbf{uIv}^T = \langle \mathbf{u}, \mathbf{v} \rangle$$

The two embeddings $\mathbf{X}_1$ and $\mathbf{X}_2$ are **equivalent** representations of the graph's latent structure. The difference arises because DeepWalk involves stochastic processes (random walks and random initialization), leading to solutions that may differ by rotation or reflection but capture the same structural information.

3 Question 3

Let us prove GNN$(PAP^T, PX) = P \cdot$ GNN(A, X).
Definitions:

- GNN Layer: GNN$(A, X) = f(\hat{A}XW)$
- Normalized Adjacency: $\hat{A} = \tilde{D}^{-\frac{1}{2}}(A + I)\tilde{D}^{-\frac{1}{2}}$

- Permutation Matrix properties: $PP^T = P^T P = I$ (Identity matrix).

First, let us define the permuted adjacency matrix as $A' = PAP^T$. We now need to find the corresponding normalized matrix $\hat{A}'$. With the normalization process being $\tilde{A} = A + I$

$$\tilde{A}' = A' + I = PAP^T + I$$

Since $I = PIP^T$, we can rewrite this as:

$$\tilde{A}' = PAP^T + PIP^T = P(A + I)P^T = P\tilde{A}P^T$$

The degree matrix $\tilde{D}$ is diagonal. When we permute the rows and columns of the adjacency matrix ($\tilde{A}' = P\tilde{A}P^T$), the corresponding degree matrix is simply the original degree matrix with its diagonal elements permuted in the same order.

$$\tilde{D}' = P\tilde{D}P^T$$

Consequently, the inverse square root also follows this transformation:

$$(\tilde{D}')^{-\frac{1}{2}} = P\tilde{D}^{-\frac{1}{2}}P^T$$

Let us now construct $\hat{A}'$:

$$\hat{A}' = (\tilde{D}')^{-\frac{1}{2}}\tilde{A}'(\tilde{D}')^{-\frac{1}{2}}$$

Substitute the permuted forms derived above and using the associative property and the orthogonality of P ($P^T P = I$):

$$\begin{aligned}
\hat{A}' &== (P\tilde{D}^{-\frac{1}{2}}P^T)(P\tilde{A}P^T)(P\tilde{D}^{-\frac{1}{2}}P^T) && (1)\\
&= P\tilde{D}^{-\frac{1}{2}}(P^T P)\tilde{A}(P^T P)\tilde{D}^{-\frac{1}{2}}P^T && (2)\\
&= P\tilde{D}^{-\frac{1}{2}}I\tilde{A}I\tilde{D}^{-\frac{1}{2}}P^T && (3)\\
&= P(\tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}})P^T && (4)\\
&= \mathbf{P\hat{A}P^T} && (5)
\end{aligned}$$

The normalized adjacency matrix permutes exactly like the original adjacency matrix.

Now, we substitute the permuted inputs A' and $X' = PX$ into the GNN equation and group the terms to isolate $P^T P$:

$$\begin{aligned}
\text{LHS} &= \text{GNN}(PAP^T, PX) && (6)\\
&= f(\hat{A}'X'W) && (7)\\
&= f((P\hat{A}P^T)(PX)W) && (8)\\
&= f(P\hat{A}(P^T P)XW) && (9)\\
&= f(P\hat{A}IXW) && (10)\\
&= f(P(\hat{A}XW)) && (11)
\end{aligned}$$

The activation function f (ReLU) is applied element-wise. Applying an element-wise function to a permuted matrix is equivalent to permuting the result of the function.

$$f(P \cdot M) = P \cdot f(M)$$

Therefore:

$$\text{LHS} = P \cdot f(\hat{A}XW)$$

Since $\text{GNN}(A, X) = f(\hat{A}XW)$, we have:

$$\boxed{\text{LHS} = \text{GNN}(PAP^T, PX) = P \cdot \text{GNN}(A, X)}$$

4 Question 4

Question 1

By definition, $\mathbf{u}$ consists of the square roots of the diagonal coefficients of $\tilde{\mathbf{D}}$, so $\mathbf{u}_i = \sqrt{\tilde{d}_i}$. The matrix $\tilde{\mathbf{D}}^{-1/2}$ is diagonal with entries $(\tilde{\mathbf{D}}^{-1/2})_{ii} = \frac{1}{\sqrt{\tilde{d}_i}}$.

Therefore, the matrix-vector multiplication yields:

$$\tilde{\mathbf{D}}^{-1/2}\mathbf{u} = \begin{bmatrix} \frac{1}{\sqrt{\tilde{d}_1}} & & 0 \\ & \ddots & \\ 0 & & \frac{1}{\sqrt{\tilde{d}_n}} \end{bmatrix} \begin{bmatrix} \sqrt{\tilde{d}_1} \\ \vdots \\ \sqrt{\tilde{d}_n} \end{bmatrix} = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} = \mathbf{1}_n$$

Moreover, we observe that multiplying the adjacency matrix by the vector of ones, $\tilde{\mathbf{A}}\mathbf{1}_n$, yields the vector of node degrees (row sums). This relationship can be expressed as:

$$\tilde{\mathbf{A}}\mathbf{1}_n = \tilde{\mathbf{D}}\mathbf{1}_n$$

Using the matrix multiplication above:

$$\begin{aligned}
\hat{\mathbf{A}}\mathbf{u} &= \left(\tilde{\mathbf{D}}^{-\frac{1}{2}}\tilde{\mathbf{A}}\tilde{\mathbf{D}}^{-\frac{1}{2}}\right)\mathbf{u} && (12)\\
&= \left(\tilde{\mathbf{D}}^{-\frac{1}{2}}\tilde{\mathbf{A}}\right)\left(\tilde{\mathbf{D}}^{-\frac{1}{2}}\mathbf{u}\right) && (13)\\
&= \tilde{\mathbf{D}}^{-\frac{1}{2}}\tilde{\mathbf{A}}\mathbf{1}_n && (14)\\
&= \tilde{\mathbf{D}}^{-\frac{1}{2}}\tilde{\mathbf{D}}\mathbf{1}_n && (15)\\
&= \tilde{\mathbf{D}}^{\frac{1}{2}}\mathbf{1}_n && (16)\\
& && (17)
\end{aligned}$$

By definition, $\tilde{\mathbf{D}}^{\frac{1}{2}}\mathbf{1}$ is exactly $\mathbf{u}$.

$$\hat{\mathbf{A}}\mathbf{u} = \mathbf{u}$$

Therefore, $\mathbf{u}$ is an eigenvector of $\hat{\mathbf{A}}$ corresponding to the eigenvalue $\lambda = 1$.

Question 2

Let the eigenvalues of $\hat{\mathbf{A}}$ be $1 = \lambda_1 > |\lambda_2| \geq \cdots \geq |\lambda_n|$ and the corresponding orthonormal eigenvectors be $v_1, v_2, \ldots, v_n$.

Note that from question 1, we know the first eigenvector corresponds to $\mathbf{u}$. Since eigenvectors in spectral decomposition must be normalized, let $\mathbf{v}_1 = \frac{\mathbf{u}}{\|\mathbf{u}\|}$.

Using spectral decomposition, we can write $\hat{\mathbf{A}}^k$ as:

$$\hat{A}^k = (\sum_{i=1}^{n} \lambda_i^k v_i v_i^T) = \lambda_1^k v_1 v_1^T + (\sum_{i=2}^{n} \lambda_i^k v_i v_i^T)$$

Thus :

$$\begin{aligned}
Z^{(k)} &= \hat{A}^k XW \\
&= \lambda_1^k v_1 v_1^T XW + (\sum_{i=2}^{n} \lambda_i^k v_i v_i^T) XW
\end{aligned}$$

Since $|\lambda_i| < 1$ for $i \geq 2$, the second term converges to 0 as $k \to \infty$. Finally :

$$\lim_{k\to\infty} Z^{(k)} = v_1 v_1^T XW = \frac{uu^T}{||u||} XW$$

Question 3

Based on the result derived in Part 2, the final representation of every node i becomes the global vector $\frac{\mathbf{u}^T\mathbf{X}\mathbf{W}}{\|\mathbf{u}\|^2}$ scaled simply by $\sqrt{\tilde{d}_i}$. Consequently, the architecture fails to distinguish between nodes because any two nodes with the same degree will necessarily have identical embeddings, causing the model to completely ignore their unique initial features $\mathbf{X}$ and their specific positions in the graph.